

ECON 0100 | Vignette A3 | Trade | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see.

Colin Creevey can bake 20 cornish pasties (P) or 5 cauldron cakes (C) in one day, and Katie Bell, back on her original oven, can bake 15 pasties or 8 cakes.

Q1 | Trade

Suppose Colin and Katie realize they can specialize and trade goods. After they specialize, what is a trade that would make them both better off?

1 C for 3 P (any rate strictly between $\frac{15}{8}$ and 4 pasties per cake is correct)

Opportunity cost of a cake: Colin 4 P , Katie $\frac{15}{8}P$. Katie has the comparative advantage in cakes and Colin in pasties, so Katie bakes cakes all day (8 C) and Colin bakes pasties all day (20 P). A price for cakes works when it sits between the two opportunity costs, $\frac{15}{8} < \text{price of 1 cake} < 4$ pasties. Below 4, Colin pays less for a cake than baking one costs him; above $\frac{15}{8}$, Katie gets more for a cake than baking one costs her. Any number in the range is correct; 2 or 3 are the natural answers.

Worked check at 1 C for 3 P , trading 2 cakes for 6 pasties. Colin ends with 14 P and 2 C ; his own PPF only allows 12 P at $C = 2$, so he consumes beyond his frontier. Katie ends with 6 C and 6 P ; her PPF allows only 3.75 P at $C = 6$, so she does too. Common wrong answers: a rate outside the range, or specializing in the wrong goods.

Q2 | Accept or Reject

Katie offers Colin 1 cake for 5 pasties. Does Colin accept? Why or why not?

Accept or Reject: Reject

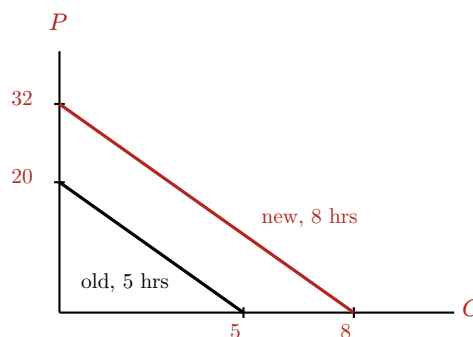
Colin's own cost of a cake is 4 pasties. Paying 5 pasties for a cake is worse than baking it himself, so he rejects; the price is above the top of the range from Q1. Katie would love this trade ($5 > \frac{15}{8}$), but a trade needs both sides. The rule students should be able to state: each party accepts only prices better than their own opportunity cost.

Q3 | Changing Labor

It turns out Colin wants to add hours to his job, so he increases from 5 to 8 hours per day. Set up Colin's old and new PPF on the same graph. What is Colin's new opportunity cost of cake? Write a short description of how this increase in hours would impact the trade with Katie you found in Q1.

New opportunity cost of 1 C : still 4 pasties

Hours scale by $\frac{8}{5}$, and so does everything Colin can make: $20 \cdot \frac{8}{5} = 32$ pasties or $5 \cdot \frac{8}{5} = 8$ cakes. The new PPF is $P = 32 - 4C$, a parallel shift out of the old one. Both intercepts grow by the same factor, so the slope is unchanged: a cake still costs Colin 4 pasties. More time changes how much he can make, not the rate at which one good trades for the other inside his own bakery.



Effect on the trade: since neither opportunity cost moved, the range of prices both accept is still $\frac{15}{8}$ to 4 pasties per cake, and the trade from Q1 still works at the same rate. What changes is the volume: Colin now has 32 pasties to work with instead of 20, so he can buy more of Katie's cakes. The size of the trade can grow; the terms do not have to. Grading note: the key idea is "opportunity cost unchanged, so the exchange rate range is unchanged." Credit "the trade gets bigger" or "Colin has more to trade" without insisting on a specific quantity.